

Objective 1: tables and figures

Machine learning and socio-cognitive traits

Scope. The main tables and figures present IV+CV results for TEA, OME and NME in the global sample (including India) and India alone. The IV-only results appear in the paired comparison table and control-increment figure. All model metrics are from the saved held-out test predictions; no model was rerun to produce this document.

Reading guide. PR-AUC is the area under the precision-recall curve; ROC-AUC is the area under the receiver-operating-characteristic curve. The positive class is outcome=1. Each feature set uses the same held-out records for a given sample and outcome.

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1 Samples and descriptive statistics

Table 1: Sample size, outcome prevalence, and data split

Sample	Target	<i>N</i>	Positive	Prev.%	Train	Valid.	Test	Countries	Years
Global	TEA	1,342,884	154,420	11.5	913,163	161,145	268,576	89	8
Global	OME	1,342,884	100,087	7.5	913,163	161,145	268,576	89	8
Global	NME	1,342,884	69,931	5.2	913,163	161,145	268,576	89	8
India	TEA	27,555	3,188	11.6	18,739	3,306	5,510	1	8
India	OME	27,555	2,167	7.9	18,739	3,306	5,510	1	8
India	NME	27,555	1,849	6.7	18,740	3,305	5,510	1	8

Note. Global includes India. The same eligible respondents and split sizes are used for IV and IV+CV within each sample and target.

Table 2: Entrepreneurial intention missingness by outcome

Sample	Target	Positive <i>N</i>	Missing	Missing%	Negative <i>N</i>	Missing	Missing%
Global	TEA	154,420	154,420	100.0	1,188,464	156,661	13.2
Global	OME	100,087	100,087	100.0	1,242,797	210,994	17.0
Global	NME	69,931	69,931	100.0	1,272,953	241,150	18.9
India	TEA	3,188	3,188	100.0	24,367	2,323	9.5
India	OME	2,167	2,167	100.0	25,388	3,344	13.2
India	NME	1,849	1,849	100.0	25,706	3,662	14.2

Note. Missing intention is retained as a distinct predictor state. Its survey pattern is strongly associated with outcome status; interpretation of model performance must account for this.

Table 3: Descriptive statistics for Objective 1 predictors

Predictor	Type	Global summary	Miss.%	India summary	Miss.%
suskill_2015_2022	Numeric	53.1% = 1	7.9	61.4% = 1	9.3
opportunity_perception	Numeric	45.9% = 1	18.1	60.4% = 1	10.4
fearfail_2015_2022	Numeric	43.7% = 1	0.6	44.0% = 1	1.0
knowing_entrepreneur	Numeric	44.2% = 1	2.3	44.8% = 1	13.1
entrepreneurial_intention	Categorical	2 categories	23.2	2 categories	20.0
media_coverage	Numeric	62.0% = 1	20.9	63.3% = 1	11.6
discent	Numeric	4.6% = 1	0.9	5.2% = 1	1.6
busang	Numeric	6.1% = 1	0.8	4.3% = 1	2.1
established_business_owner	Numeric	7.8% = 1	0.0	6.6% = 1	0.0
good_career_choice	Numeric	1.25 (0.93)	14.3	1.32 (0.91)	6.5
ease_start_business	Numeric	43.7% = 1	22.2	59.2% = 1	11.8
high_status	Numeric	69.8% = 1	22.8	70.6% = 1	11.6
social_enterprise	Numeric	39.4% = 1	32.4	58.0% = 1	13.2
yrsurv	Categorical	8 categories	0.0	8 categories	0.0
age	Numeric	41.50 (14.53)	2.8	35.71 (12.20)	0.0
gender	Categorical	2 categories	0.0	2 categories	0.0
education_attainment	Categorical	5 categories	1.6	5 categories	0.5
household_income	Categorical	3 categories	18.9	3 categories	6.1
hhsiz	Numeric	3 [2, 4]	1.8	4 [3, 5]	1.1
occupation_status	Categorical	7 categories	1.8	7 categories	4.5
gdp_per_capita_ppp	Numeric	35,593 (21,375)	0.0	6,357 (857)	0.0
population_totalthousands	Numeric	96,167,410 (259,900,717)	0.0	1,363,227,202 (32,494,679)	0.0
gdp_growth	Numeric	2.41 (4.02)	0.0	5.63 (4.43)	0.0

Note. Binary variables show the observed percentage coded 1. Other numeric variables show mean (SD), except household size, which shows median [Q1, Q3] because of extreme values. Categorical variables show the number of observed categories. Values are respondent-weighted. Missingness uses all eligible respondents as denominator. Survey year and country-level variables are controls.

2 Classifier results

Table 4: Held-out classifier performance, IV+CV

Model	ROC-AUC	PR-AUC	Prec.	Recall	F1	Acc.	Thresh.	Test N
Global TEA test N=268,576								
Logistic Regression	0.991	0.929	0.843	0.885	0.864	0.968	0.500	268,576
Lasso	0.991	0.929	0.846	0.881	0.863	0.968	0.510	268,576
Ridge	0.991	0.927	0.844	0.877	0.860	0.967	0.452	268,576
Elastic Net	0.991	0.929	0.846	0.881	0.863	0.968	0.509	268,576
Naive Bayes	0.939	0.586	0.596	0.833	0.695	0.916	0.434	268,576
Decision Tree	0.989	0.892	0.845	0.871	0.858	0.967	0.500	268,576
XGBoost	0.993	0.950	0.878	0.888	0.883	0.973	0.500	268,576
CatBoost	0.993	0.947	0.861	0.899	0.880	0.972	0.456	268,576
Random Forest	0.993	0.950	0.865	0.903	0.884	0.973	0.441	268,576
H2O GBM	0.992	0.936	0.849	0.888	0.868	0.969	0.505	268,576
Global OME test N=268,576								
Logistic Regression	0.974	0.677	0.578	0.843	0.686	0.942	0.325	268,576
Lasso	0.974	0.678	0.578	0.843	0.686	0.942	0.328	268,576
Ridge	0.973	0.672	0.580	0.843	0.687	0.943	0.298	268,576
Elastic Net	0.974	0.678	0.578	0.843	0.686	0.942	0.328	268,576
Naive Bayes	0.920	0.417	0.417	0.777	0.543	0.902	1.000	268,576
Decision Tree	0.971	0.642	0.559	0.844	0.672	0.939	0.347	268,576
XGBoost	0.978	0.731	0.623	0.825	0.710	0.950	0.364	268,576
CatBoost	0.977	0.724	0.595	0.865	0.705	0.946	0.314	268,576
Random Forest	0.978	0.738	0.624	0.827	0.712	0.950	0.354	268,576
H2O GBM	0.976	0.702	0.586	0.855	0.695	0.944	0.337	268,576
Global NME test N=268,576								
Logistic Regression	0.974	0.647	0.539	0.715	0.614	0.953	0.311	268,576
Lasso	0.974	0.648	0.539	0.716	0.615	0.953	0.313	268,576
Ridge	0.973	0.632	0.572	0.651	0.609	0.957	0.316	268,576
Elastic Net	0.974	0.648	0.539	0.716	0.615	0.953	0.312	268,576
Naive Bayes	0.924	0.341	0.339	0.712	0.460	0.913	0.714	268,576
Decision Tree	0.968	0.588	0.669	0.523	0.587	0.962	0.440	268,576
XGBoost	0.979	0.709	0.613	0.707	0.657	0.962	0.335	268,576
CatBoost	0.979	0.706	0.612	0.704	0.655	0.961	0.337	268,576
Random Forest	0.979	0.714	0.621	0.711	0.663	0.962	0.325	268,576
H2O GBM	0.977	0.683	0.600	0.691	0.642	0.960	0.338	268,576
India TEA test N=5,510								
Logistic Regression	0.997	0.977	0.928	0.948	0.938	0.985	0.606	5,510
Lasso	0.997	0.975	0.918	0.948	0.933	0.984	0.640	5,510
Ridge	0.997	0.975	0.922	0.961	0.941	0.986	0.500	5,510

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Table continued

Model	ROC-AUC	PR-AUC	Prec.	Recall	F1	Acc.	Thresh.	Test N
Elastic Net	0.997	0.975	0.921	0.950	0.935	0.985	0.636	5,510
Naive Bayes	0.938	0.680	0.649	0.826	0.727	0.928	0.211	5,510
Decision Tree	0.994	0.920	0.916	0.956	0.935	0.985	0.500	5,510
XGBoost	0.997	0.977	0.913	0.961	0.936	0.985	0.500	5,510
CatBoost	0.997	0.976	0.927	0.956	0.941	0.986	0.614	5,510
Random Forest	0.997	0.979	0.930	0.954	0.942	0.986	0.474	5,510
H2O GBM	0.997	0.978	0.934	0.950	0.942	0.986	0.630	5,510
India OME test N=5,510								
Logistic Regression	0.987	0.844	0.706	0.898	0.791	0.963	0.373	5,510
Lasso	0.987	0.842	0.696	0.894	0.783	0.961	0.403	5,510
Ridge	0.987	0.844	0.707	0.891	0.789	0.962	0.347	5,510
Elastic Net	0.986	0.839	0.729	0.820	0.772	0.962	0.473	5,510
Naive Bayes	0.913	0.574	0.546	0.661	0.598	0.930	0.905	5,510
Decision Tree	0.984	0.789	0.678	0.845	0.752	0.956	0.296	5,510
XGBoost	0.986	0.832	0.697	0.889	0.782	0.961	0.337	5,510
CatBoost	0.988	0.854	0.706	0.878	0.783	0.962	0.403	5,510
Random Forest	0.988	0.856	0.744	0.820	0.780	0.964	0.368	5,510
H2O GBM	0.988	0.851	0.743	0.801	0.771	0.963	0.495	5,510
India NME test N=5,510								
Logistic Regression	0.986	0.803	0.739	0.775	0.757	0.967	0.459	5,510
Lasso	0.987	0.805	0.733	0.772	0.752	0.966	0.466	5,510
Ridge	0.986	0.805	0.738	0.778	0.757	0.967	0.390	5,510
Elastic Net	0.987	0.805	0.735	0.775	0.755	0.966	0.464	5,510
Naive Bayes	0.904	0.429	0.496	0.607	0.546	0.932	0.974	5,510
Decision Tree	0.986	0.765	0.763	0.732	0.747	0.967	0.500	5,510
XGBoost	0.988	0.826	0.761	0.775	0.768	0.969	0.468	5,510
CatBoost	0.988	0.829	0.758	0.772	0.765	0.968	0.467	5,510
Random Forest	0.987	0.818	0.691	0.837	0.757	0.964	0.310	5,510
H2O GBM	0.988	0.823	0.699	0.829	0.758	0.965	0.380	5,510

Note. Each group uses the same test respondents across models. Thresholds were selected from validation data; metrics are evaluated on held-out test data. Scores are rounded for display; source CSVs retain full precision.

Table 5: Held-out confusion matrices, IV+CV

Model	TP	FP	TN	FN	Test N	Thresh.
Global TEA						
Logistic Regression	27,345	5,099	232,593	3,539	268,576	0.500
Lasso	27,213	4,958	232,734	3,671	268,576	0.510
Ridge	27,097	5,002	232,690	3,787	268,576	0.452
Elastic Net	27,216	4,961	232,731	3,668	268,576	0.509
Naive Bayes	25,723	17,468	220,224	5,161	268,576	0.434
Decision Tree	26,898	4,952	232,740	3,986	268,576	0.500
XGBoost	27,415	3,805	233,887	3,469	268,576	0.500
CatBoost	27,770	4,476	233,216	3,114	268,576	0.456
Random Forest	27,897	4,367	233,325	2,987	268,576	0.441
H2O GBM	27,424	4,886	232,806	3,460	268,576	0.505
Global OME						
Logistic Regression	16,877	12,305	236,254	3,140	268,576	0.325
Lasso	16,871	12,329	236,230	3,146	268,576	0.328
Ridge	16,868	12,234	236,325	3,149	268,576	0.298
Elastic Net	16,867	12,325	236,234	3,150	268,576	0.328
Naive Bayes	15,545	21,738	226,821	4,472	268,576	1.000
Decision Tree	16,894	13,334	235,225	3,123	268,576	0.347
XGBoost	16,512	10,008	238,551	3,505	268,576	0.364
CatBoost	17,315	11,778	236,781	2,702	268,576	0.314
Random Forest	16,562	9,975	238,584	3,455	268,576	0.354
H2O GBM	17,121	12,120	236,439	2,896	268,576	0.337
Global NME						
Logistic Regression	9,993	8,550	246,040	3,993	268,576	0.311
Lasso	10,016	8,574	246,016	3,970	268,576	0.313
Ridge	9,111	6,807	247,783	4,875	268,576	0.316
Elastic Net	10,020	8,574	246,016	3,966	268,576	0.312
Naive Bayes	9,958	19,381	235,209	4,028	268,576	0.714
Decision Tree	7,311	3,610	250,980	6,675	268,576	0.440
XGBoost	9,893	6,239	248,351	4,093	268,576	0.335
CatBoost	9,845	6,238	248,352	4,141	268,576	0.337
Random Forest	9,949	6,080	248,510	4,037	268,576	0.325
H2O GBM	9,668	6,449	248,141	4,318	268,576	0.338
India TEA						
Logistic Regression	604	47	4,826	33	5,510	0.606
Lasso	604	54	4,819	33	5,510	0.640
Ridge	612	52	4,821	25	5,510	0.500
Elastic Net	605	52	4,821	32	5,510	0.636
Naive Bayes	526	285	4,588	111	5,510	0.211
Decision Tree	609	56	4,817	28	5,510	0.500
XGBoost	612	58	4,815	25	5,510	0.500
CatBoost	609	48	4,825	28	5,510	0.614
Random Forest	608	46	4,827	29	5,510	0.474
H2O GBM	605	43	4,830	32	5,510	0.630
India OME						
Logistic Regression	389	162	4,915	44	5,510	0.373
Lasso	387	169	4,908	46	5,510	0.403
Ridge	386	160	4,917	47	5,510	0.347
Elastic Net	355	132	4,945	78	5,510	0.473
Naive Bayes	286	238	4,839	147	5,510	0.905
Decision Tree	366	174	4,903	67	5,510	0.296
XGBoost	385	167	4,910	48	5,510	0.337
CatBoost	380	158	4,919	53	5,510	0.403
Random Forest	355	122	4,955	78	5,510	0.368
H2O GBM	347	120	4,957	86	5,510	0.495
India NME						
Logistic Regression	286	101	5,040	83	5,510	0.459
Lasso	285	104	5,037	84	5,510	0.466
Ridge	287	102	5,039	82	5,510	0.390
Elastic Net	286	103	5,038	83	5,510	0.464
Naive Bayes	224	228	4,913	145	5,510	0.974
Decision Tree	270	84	5,057	99	5,510	0.500
XGBoost	286	90	5,051	83	5,510	0.468
CatBoost	285	91	5,050	84	5,510	0.467
Random Forest	309	138	5,003	60	5,510	0.310
H2O GBM	306	132	5,009	63	5,510	0.380

Note. TP=true positives; FP=false positives; TN=true negatives; FN=false negatives. Counts use each model's validation-selected threshold.

3 Effect of adding controls

Table 6: IV versus IV+CV on matched held-out respondents

Model	PR IV	PR +CV	Δ PR	Δ ROC	Δ F1	F1 IV	F1 +CV
Global TEA							
Logistic Regression	0.883	0.929	+0.046	+0.003	+0.020	0.844	0.864
Lasso	0.883	0.929	+0.046	+0.004	+0.019	0.844	0.863
Ridge	0.882	0.927	+0.045	+0.003	+0.018	0.842	0.860
Elastic Net	0.883	0.929	+0.046	+0.004	+0.020	0.843	0.863
Naive Bayes	0.558	0.586	+0.028	+0.003	+0.015	0.680	0.695
Decision Tree	0.814	0.892	+0.078	+0.005	+0.013	0.845	0.858
XGBoost	0.891	0.950	+0.059	+0.005	+0.034	0.849	0.883
CatBoost	0.890	0.947	+0.057	+0.005	+0.031	0.849	0.880
Random Forest	0.889	0.950	+0.061	+0.005	+0.035	0.849	0.884
H2O GBM	0.885	0.936	+0.051	+0.004	+0.021	0.847	0.868
Global OME							
Logistic Regression	0.634	0.677	+0.044	+0.003	+0.017	0.669	0.686
Lasso	0.634	0.678	+0.044	+0.003	+0.017	0.668	0.686
Ridge	0.632	0.672	+0.040	+0.003	+0.019	0.667	0.687
Elastic Net	0.634	0.678	+0.044	+0.003	+0.017	0.668	0.686
Naive Bayes	0.398	0.417	+0.019	+0.001	+0.010	0.532	0.543
Decision Tree	0.594	0.642	+0.048	+0.003	+0.014	0.658	0.672
XGBoost	0.646	0.731	+0.085	+0.006	+0.036	0.674	0.710
CatBoost	0.646	0.724	+0.078	+0.006	+0.032	0.674	0.705
Random Forest	0.645	0.738	+0.093	+0.007	+0.038	0.673	0.712
H2O GBM	0.642	0.702	+0.060	+0.005	+0.024	0.671	0.695
Global NME							
Logistic Regression	0.461	0.647	+0.187	+0.014	+0.083	0.532	0.614
Lasso	0.461	0.648	+0.187	+0.014	+0.083	0.532	0.615
Ridge	0.454	0.632	+0.177	+0.013	+0.078	0.531	0.609
Elastic Net	0.461	0.648	+0.187	+0.014	+0.083	0.532	0.615
Naive Bayes	0.301	0.341	+0.041	+0.010	+0.032	0.428	0.460
Decision Tree	0.385	0.588	+0.203	+0.015	+0.069	0.518	0.587
XGBoost	0.489	0.709	+0.221	+0.016	+0.117	0.539	0.657
CatBoost	0.488	0.706	+0.218	+0.016	+0.116	0.539	0.655
Random Forest	0.488	0.714	+0.226	+0.017	+0.125	0.538	0.663
H2O GBM	0.476	0.683	+0.207	+0.015	+0.106	0.537	0.642
India TEA							
Logistic Regression	0.970	0.977	+0.007	+0.001	+0.008	0.930	0.938
Lasso	0.970	0.975	+0.005	+0.000	-0.000	0.933	0.933

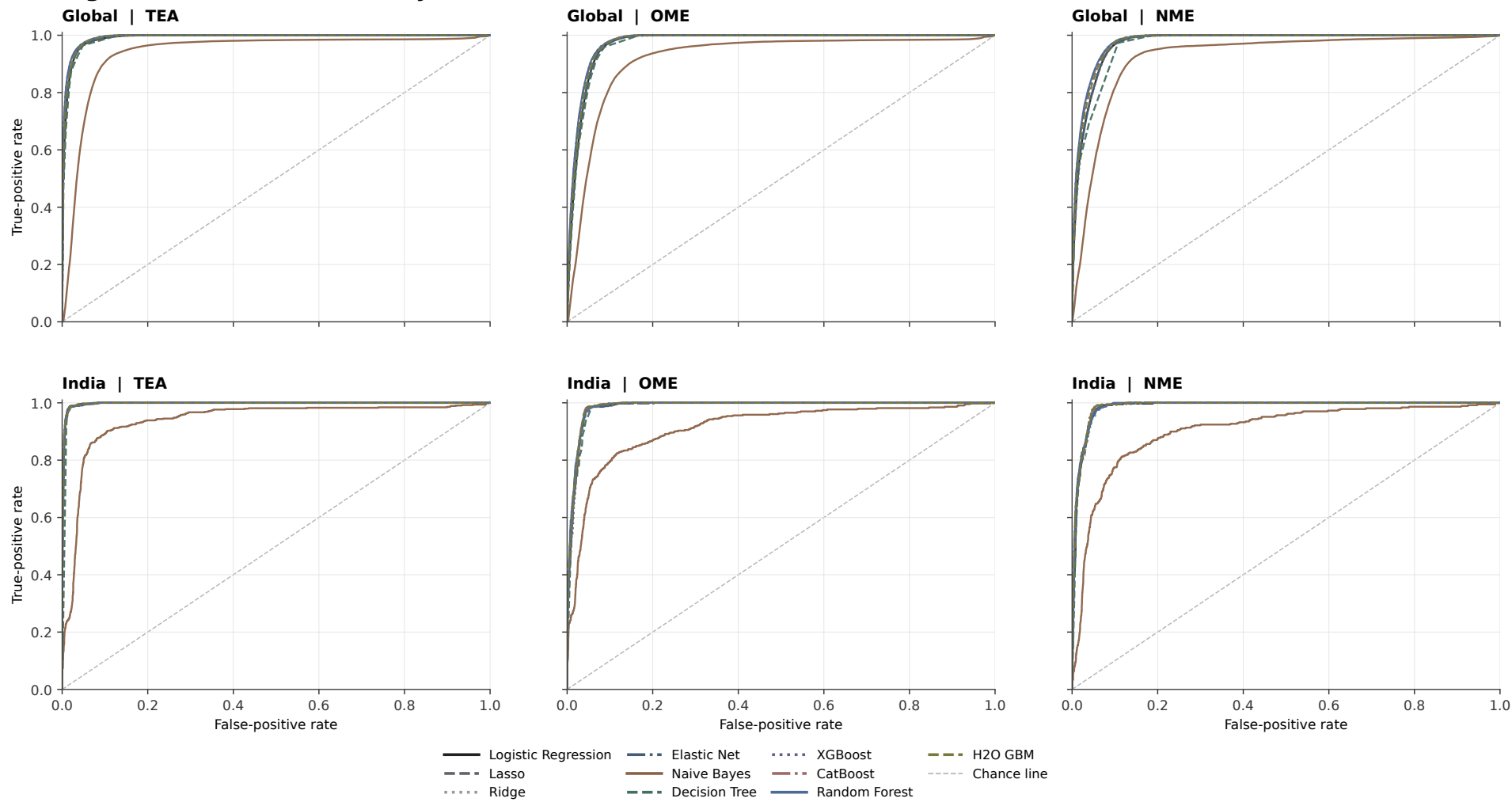
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Table continued							
Model	PR IV	PR +CV	Δ PR	Δ ROC	Δ F1	F1 IV	F1 +CV
Ridge	0.967	0.975	+0.008	+0.001	+0.008	0.933	0.941
Elastic Net	0.970	0.975	+0.006	+0.000	+0.002	0.933	0.935
Naive Bayes	0.679	0.680	+0.000	+0.003	-0.014	0.740	0.727
Decision Tree	0.921	0.920	-0.001	-0.000	+0.001	0.935	0.935
XGBoost	0.968	0.977	+0.009	+0.001	+0.005	0.931	0.936
CatBoost	0.970	0.976	+0.007	+0.001	+0.011	0.930	0.941
Random Forest	0.973	0.979	+0.006	+0.000	+0.010	0.932	0.942
H2O GBM	0.968	0.978	+0.010	+0.001	+0.008	0.934	0.942
India OME							
Logistic Regression	0.805	0.844	+0.039	+0.002	+0.019	0.771	0.791
Lasso	0.804	0.842	+0.038	+0.002	+0.032	0.751	0.783
Ridge	0.805	0.844	+0.038	+0.002	+0.019	0.769	0.789
Elastic Net	0.802	0.839	+0.037	+0.002	+0.015	0.757	0.772
Naive Bayes	0.573	0.574	+0.001	+0.006	-0.015	0.613	0.598
Decision Tree	0.754	0.789	+0.035	+0.002	+0.010	0.743	0.752
XGBoost	0.801	0.832	+0.030	+0.002	+0.016	0.766	0.782
CatBoost	0.807	0.854	+0.047	+0.003	+0.031	0.751	0.783
Random Forest	0.814	0.856	+0.043	+0.003	+0.008	0.772	0.780
H2O GBM	0.805	0.851	+0.046	+0.003	+0.001	0.770	0.771
India NME							
Logistic Regression	0.639	0.803	+0.164	+0.009	+0.073	0.683	0.757
Lasso	0.634	0.805	+0.171	+0.010	+0.056	0.696	0.752
Ridge	0.652	0.805	+0.153	+0.009	+0.077	0.681	0.757
Elastic Net	0.634	0.805	+0.171	+0.010	+0.064	0.690	0.755
Naive Bayes	0.415	0.429	+0.014	-0.006	-0.019	0.565	0.546
Decision Tree	0.622	0.765	+0.143	+0.009	+0.049	0.697	0.747
XGBoost	0.679	0.826	+0.146	+0.008	+0.048	0.720	0.768
CatBoost	0.664	0.829	+0.165	+0.009	+0.055	0.710	0.765
Random Forest	0.682	0.818	+0.135	+0.007	+0.039	0.718	0.757
H2O GBM	0.677	0.823	+0.146	+0.008	+0.036	0.722	0.758

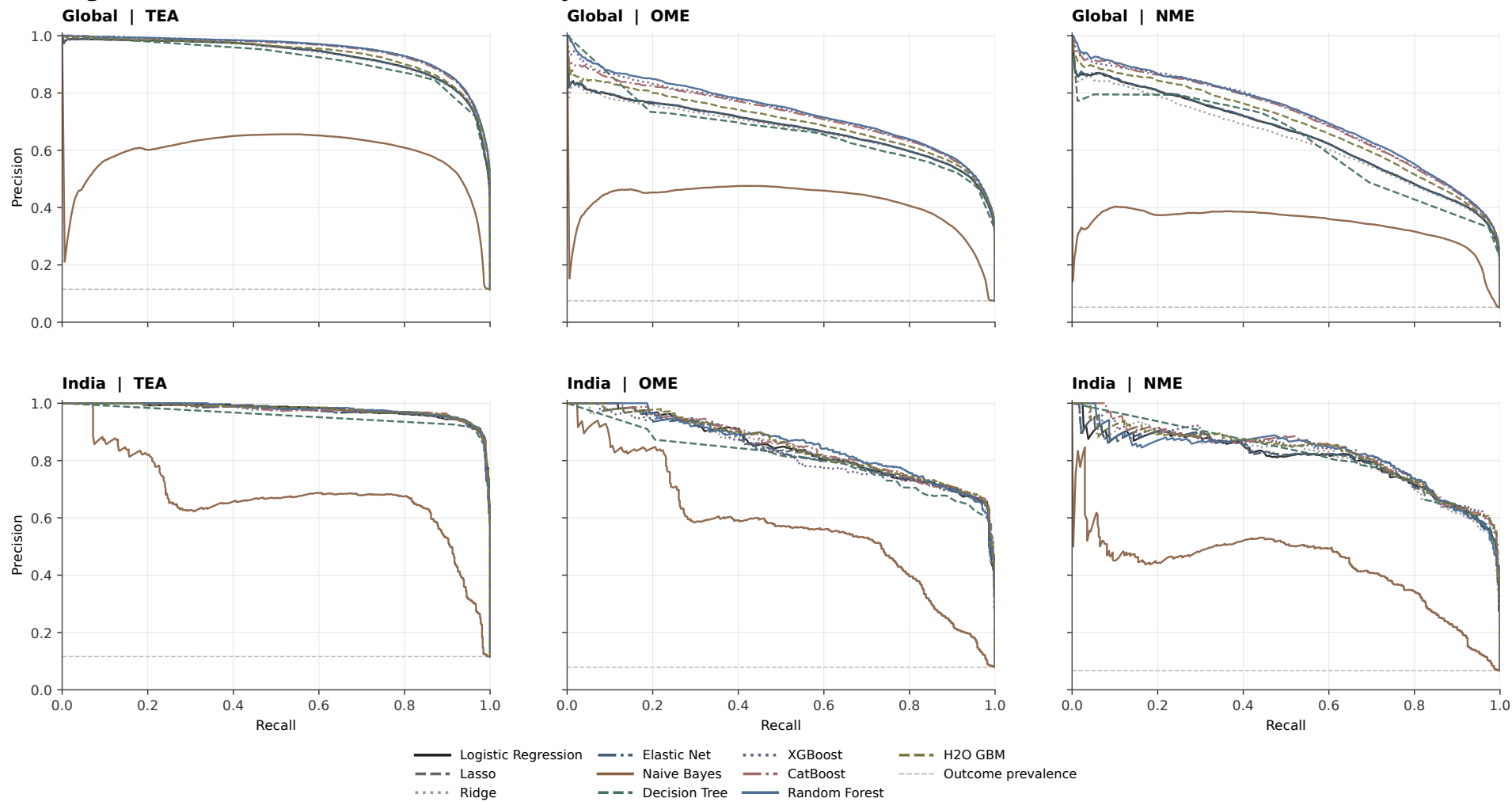
Note. Delta is IV+CV minus IV. Held-out row IDs were checked and match exactly for every sample-target pair. No statistical significance is claimed by these descriptive differences.

4 Figures

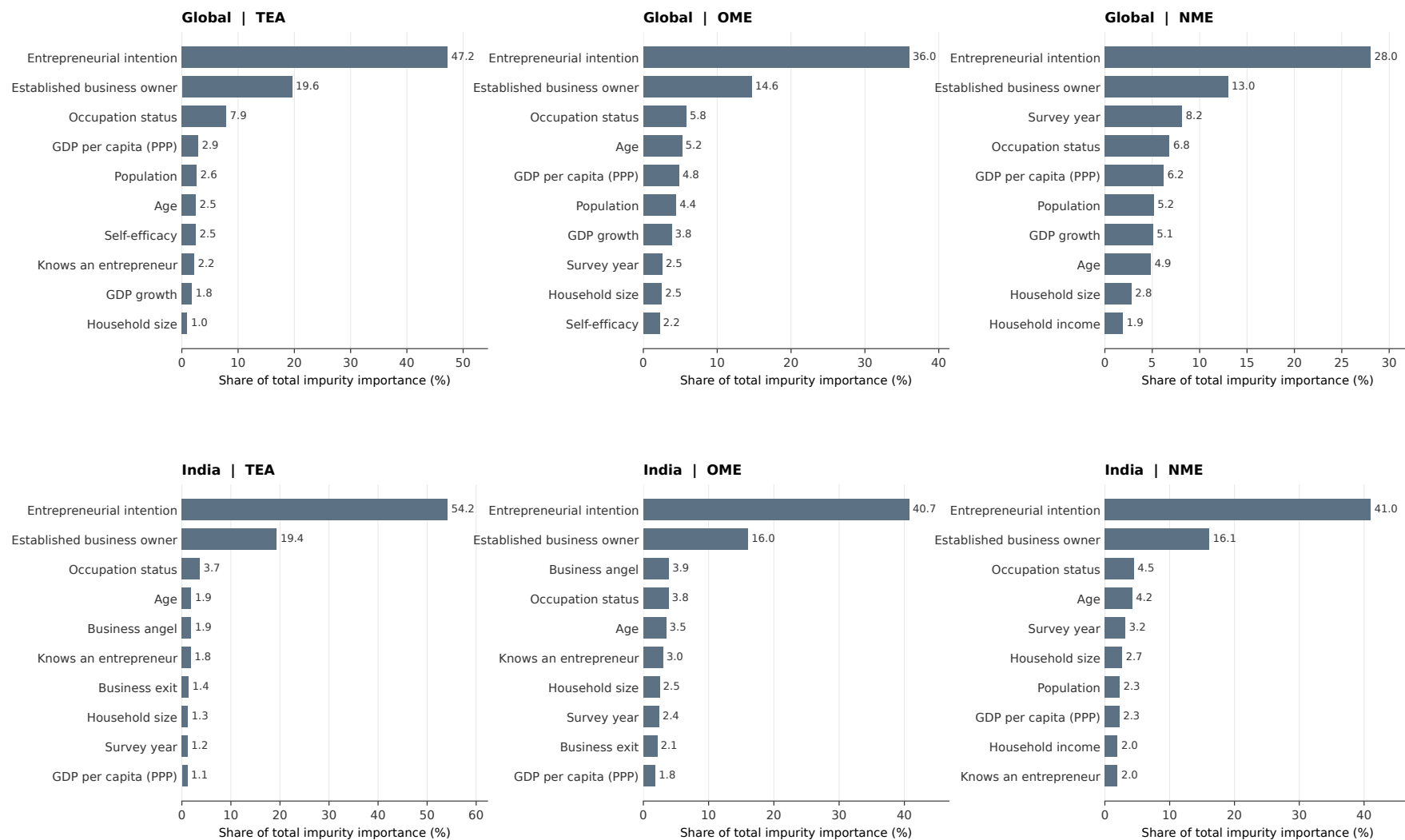
Figure 1. ROC curves for Objective 1



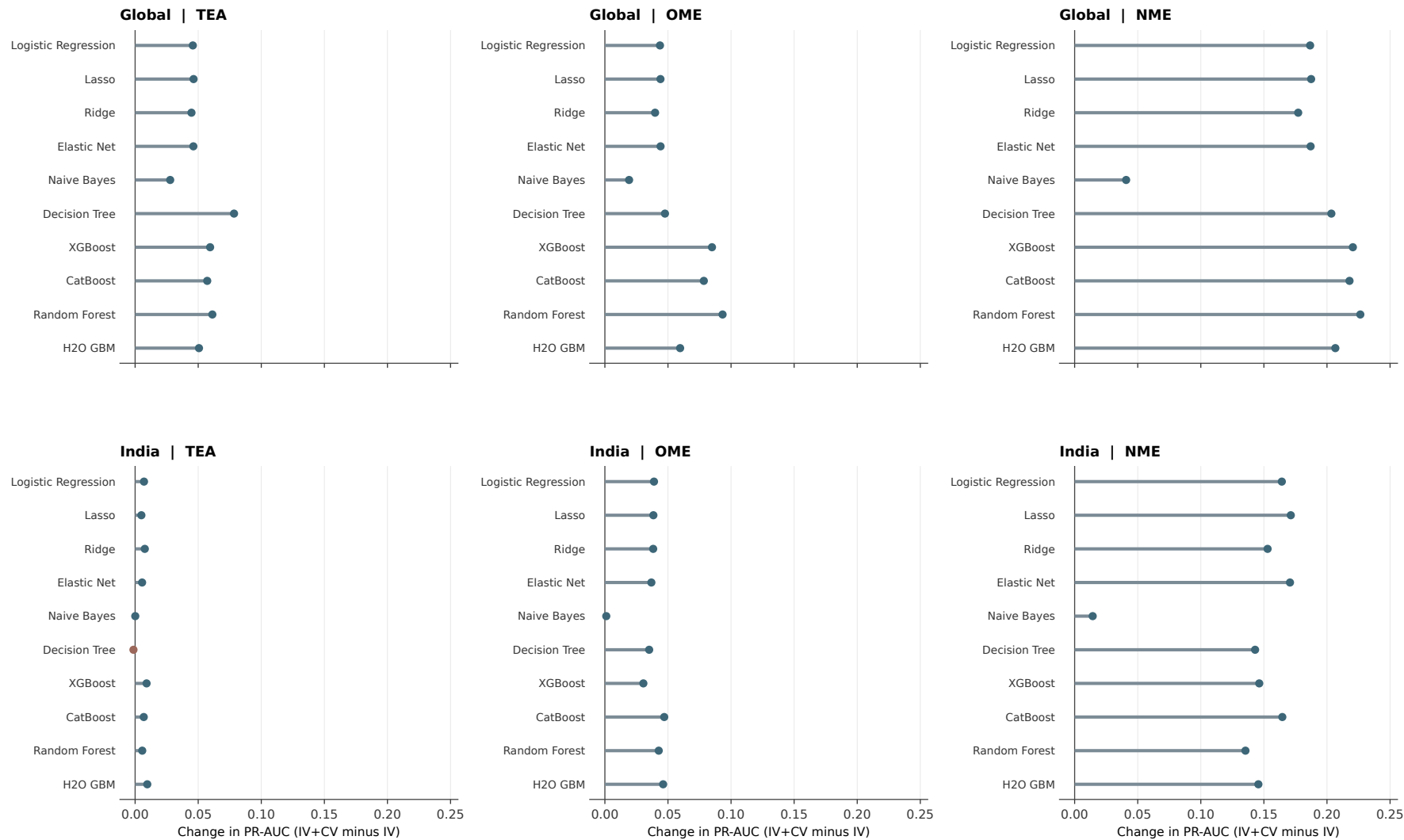
IV+CV models; held-out test data. Global includes India. All models in each panel use the same test records.

Figure 2. Precision-recall curves for Objective 1

IV+CV models; held-out test data. Global includes India. All models in each panel use the same test records. Dashed lines show outcome prevalence.

Figure 3. Random-forest feature importance for Objective 1

IV+CV models. Top 10 features per panel; percentages use all fitted features as denominator. Impurity importance is descriptive and does not establish causal effects. Entrepreneurial intention has outcome-linked missingness (Table 2).

Figure 4. Predictive change after adding controls

Positive values favour IV+CV. Each pair uses exactly the same held-out respondents. Differences describe this test split; no causal effect or significance test is implied.

A Variable and model reference

Table A1: Description and type of variables used in Objective 1

Variable	Role	Description	Type
country_name	sample selector	Survey country; selects the Global or India sample.	character
setid	record identifier	Respondent or observation identifier.	numeric
TEA	outcome	Indicator of total early-stage entrepreneurial activity.	numeric
OME	outcome	Indicator of opportunity-motivated early-stage entrepreneurship.	numeric
NME	outcome	Indicator of necessity-motivated early-stage entrepreneurship.	numeric
suskill_2015_2022	independent variable	Perceived knowledge, skills and experience to start a business.	numeric
opportunity_perception	independent variable	Perception of good opportunities to start a business.	numeric
fearfail_2015_2022	independent variable	Fear of failure as a barrier to starting a business.	numeric
knowing_entrepreneur	independent variable	Personally knows someone who started a business.	numeric
entrepreneurial_intention	independent variable	Intention to start a business among eligible non-entrepreneurs.	numeric
media_coverage	independent variable	Perceived media coverage of successful new businesses.	numeric
discent	independent variable	Reported business discontinuation or exit.	numeric
busang	independent variable	Reported business angel investment.	numeric
established_business_owner	independent variable	Ownership or management of an established business.	numeric
good_career_choice	independent variable	Belief that entrepreneurship is a desirable career choice.	numeric
ease_start_business	independent variable	Perceived ease of starting a business.	numeric
high_status	independent variable	Belief that successful entrepreneurs have high status.	numeric
social_enterprise	independent variable	Social or environmental purpose of entrepreneurial activity.	numeric
yrsurv	control	Year of the GEM survey.	integer
age	control	Respondent age at survey.	numeric
gender	control	Respondent gender code.	numeric
education_attainment	control	Harmonized educational attainment category.	numeric
household_income	control	Relative household income category.	numeric
hhsiz	control	Number of people in the respondent's household.	numeric
occupation_status	control	Respondent occupational status category.	numeric
gdp_per_capita_ppp	control	Country GDP per capita adjusted for purchasing power.	numeric
population_totalthousands	control	Country population measure as stored in the source data.	integer
gdp_growth	control	Annual country GDP growth rate.	numeric

Note. Country identifies the sample and setid identifies the record; neither enters a model. Descriptions are drawn from the supplied variable dictionary.

Table A2: Advantages, limitations, and observed model runtime

Model	Model type	Advantage	Limitation	Min.
Logistic Regression	Logistic probability model with a small fixed stabilizing penalty	Clear probability baseline	Small penalty is needed because intention missingness separates classes	7.0
Lasso	Logistic model with L1 shrinkage	Can select predictors	May select one of several correlated predictors	45.6
Ridge	Logistic model with L2 shrinkage	Stable with correlated predictors	Does not select predictors	39.7
Elastic Net	Logistic model with mixed L1 and L2 shrinkage	Can select predictors while handling correlation	Requires tuning two penalty components	52.6
Naive Bayes	Probabilistic classifier using conditional feature distributions	Fast baseline	Assumes conditional feature independence	15.1
Decision Tree	Single classification tree	Simple to inspect	Can be unstable and less accurate	0.7
XGBoost	Gradient boosted trees	Strong nonlinear benchmark	Requires tuning and more compute	1.8
CatBoost	Gradient boosted trees with native categorical handling	Handles categorical predictors	Requires separate installation and more compute	11.8
Random Forest	Ensemble of randomized decision trees	Strong nonlinear benchmark	Requires more memory and compute	10.3
H2O GBM	H2O gradient boosted trees	Scalable distributed implementation	Requires Java and a local H2O service	3.7

Note. Runtime is total fit-and-predict time across all 12 Objective 1 cases, not time per case.